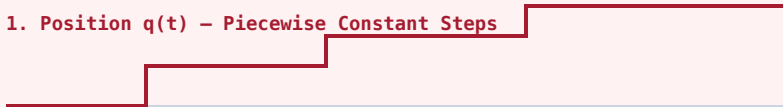


# DERIVATIVE CONTINUITY (C<sup>2</sup>) VS. DISCONTINUOUS WAYPOINT STEPPING

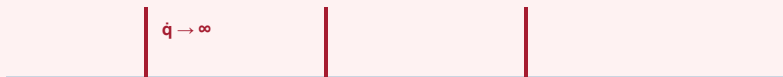
How quintic polynomial spline interpolation bounds acceleration and eliminates infinite jerk spikes ( $\delta(t)$ )

## ✗ DISCONTINUOUS C0 STEPPING (ZERO-ORDER HOLD)

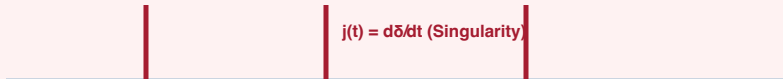
### 1. Position $q(t)$ – Piecewise Constant Steps



### 2. Velocity $\dot{q}(t)$ – Infinite Dirac Spikes $\delta(t)$



### 3. Jerk $j(t) = d^3q/dt^3$ – Unbounded Chatter



#### PHYSICAL HARDWARE CONSEQUENCES:

- **Extreme Torque Spikes:** Motor commanded to accelerate instantly.
- **Current Inrush ( $I^2 R$ ):** Massive coil heating, blowing power fuses.
- **Gearbox Backlash & Wear:** Strips reduction gear teeth.
- **Audio Acoustics:** Loud, high-pitched mechanical knocking.

## ✓ C2 CONTINUITY (QUINTIC POLYNOMIAL SPLINE)

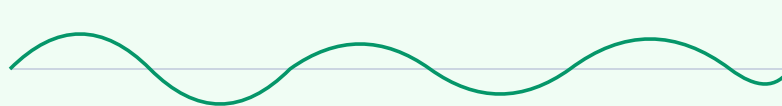
### 1. Position $q(t)$ – Smooth Quintic Interpolation



### 2. Velocity $\dot{q}(t)$ – Continuous & Bounded ( $|\dot{q}| \leq v_{\max}$ )



### 3. Jerk $j(t) = d^3q/dt^3$ – Bounded & Smooth



#### PHYSICAL SYSTEMS ADVANTAGES:

- **Smooth Motor Commutation:** Current follows gradual sinusoidal curves.
- **Thermal Protection:** Operates within nominal continuous torque ratings.
- **10x Extended Lifespan:** Zero harmonic resonance or gearbox damage.
- **Rock-Solid Tracking:** 1 kHz MCU PID loops track trajectory perfectly.